Optimizing Automotive Seating Position to Reduce Injury Severity in Short-Statured Drivers

Optimal Seating Position Verification System

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Table of Contents

Table of Contents **1**
Introduction **3**
Project Goal 3
Background 3
Injury Statistics 3
Electronics 4
Previous Year's Research Summary **5**
Driver Seat: Relative Position Measurements Diagram1 *6*
Composite Injury Score Guidelines *6*
Simulation Data Results Table *7*
Participant 15 As Baselined in LS-DYNA *7*
Participant 15 As Baselined Post-Processing Results *7*
Participant 15 As Optimized in LS-DYNA *8*
Participant 15 As Optimized Post-Processing Results *8*
Comparison of Participant 15 Measurements Table *8*
Composite Injury Score Plot *9*
Requirements **9**
OSPVS Conceptual Block Definition Diagram generated in Lucid *9*
Procedure **10**
OSPVS Procedure Flow Chart generated in Cameo Enterprise Architecture *10*
Data Results and Analysis **11**
Height vs Chest Distance Regression 11
Height vs Chest Distance (Optimized) Regression generated in Excel *11*
Height vs Chest Distance (Baselined) Regression generated in Excel *12*
Initial Iteration 13
Sketch 1: Initial Visuino Sketch with One ESP32 and Two Sensors *13*
Current Visuino Sketches 14
Sketch 2: Current Visuino Sketch for Pelvic Angle Collection *14*
Sketch 3: Current Visuino Sketch for Chest Distance Collection *15*
Physical Prototype 16
Physical Prototype: Chest Device on Left, Pelvic Angle Device on Right *16*
Adafruit Dashboard 17
Adafruit Dashboard Image *17*
Error Analysis **18**
Applications and Future Plans **18**
References **18**
Acknowledgements and Statement on Outside Assistance **19**
Abstract **20**

Introduction

Project Goal

The previous year's goal was to find the optimal seating position for short-statured drivers that reduces the severity of injury in a frontal collision by performing a multi-physics, in-silica simulation using LS-DYNA to determine force and duration of impact through a range of possible seating positions for drivers of varying stature/anatomy and compare the force to the maximum force the human body can handle depending on their anatomy.

This year, the goal is to leverage last year's conclusions regarding seating position to create a system that will alert a driver if they are in their proper position based on stature and prompt readjustment if they are not. Sensors will collect the relevant distance and rotational measurements and it will be run through an algorithm that will take into account the driver's measurements, determine their ideal position, and check if each measurement is within a proper and safe range.

Background

The student researcher was inspired by her own experience. When she began doing her in-car driver's education in summer 2019, she found that there was no truly safe position for her to drive in as a 5'2" female. If she sat in the car's default driving position, she couldn't reach the pedals or see over the steering wheel, while if she adjusted her seat to where she could properly see and reach everything, her head was a foot or less away from the windshield. After looking into this, the student researcher realized that this was not, in fact, an isolated issue, nor her being far too anxious. This is an actual issue.

Cars are a necessary aspect of our modern lives. According to the American Driving Survey (2015-2016) conducted by the Foundation for Traffic Safety, people spend an average of nearly one hour behind the wheel per day.[7] That hour provides countless opportunities to have your life gravely affected, and this disproportionately affects short-statured people, who generally happen to be women.

Injury Statistics

Automotive safety including both driver and passenger injuries has been heavily studied in the last three decades, and have produced fascinating discoveries. Short-statured drivers are 47% more likely to be severely injured and 17% more likely to die in a car crash. This discrepancy is due to insufficient vehicle safety guidelines resulting from continued use of physical crash test dummies with limited biofidelity. A prime example of this is the fact that female crash test dummies were introduced 60 years after male dummies, and none of these dummies are anthropomorphically correct.Even today, no testing requires accurate dummies that

are representative of the broader population. Most testing is done with a 50th percentile male dummy.[10]

While testing for the average of the population is logical, it endangers the extreme ends of the population. Simply extrapolating or adjusting data is not accurate.This places drivers up to 155 cm tall (short-statured) at a greater risk of serious head, pelvis, and lower extremity injury.[4]

One of the most important factors in determining head acceleration and injury is distance from the airbag to the dummy/person. Increasing airbag distance decreases injury, as indicated by chest deflection or Head Injury Criteria. Shorter drivers typically position themselves much closer to the airbag.[11]

Electronics

The main electronic components used in this project are microcontrollers, ultrasonic range finders, and an accelerometer.

Microcontrollers are compact integrated circuits designed to perform specific tasks. Microcontrollers typically combine a processor, read-only memory, and input-output functions on a single chip or a small printed circuit board. Typical applications for microcontrollers include embedded systems that are part of manufacturing systems, robotics, automobiles, home appliances, office machines, medical and life support systems, and security systems, as well as internet of things (IoT) implementations.[12]

Specifically, Arduino microcontrollers were selected as the basis for this project. Arduinos are small, inexpensive, open-source, have a visual design platform (Visuino), and offer a broad selection of compatible sensors. On top of this, the student researcher already had experience with Arduinos from previous research.

While most microcontrollers are programmed utilizing the C coding language, the Arduino development environment is designed to hide most of the complexity from the programmer as it is designed for beginners.

The specific microcontroller used is called an ESP32, selected because it has WiFi and Bluetooth capabilities, which would make for easy transfer of data.[13] The microcontroller connects to and communicates with the connected sensors, an ultrasonic rangefinder and a gyroscope. An ultrasonic range finder has two main components: ping, which sends out the sound waves, and echo, which hears the sound waves. The HC-SR04 ultrasonic range finder used emits sound waves beyond the audible human range of 20 Hz to 20 kHz.[14]The pelvic angle is collected by an MPU9250 accelerometer, which is run through an acceleration to angle function inside of Visuino.

The other main function in Visuino is MQTT (Message Queue Telemetry Transport), which is an IoT (Internet of Things) messaging protocol. It works in conjunction with the ESP32's WiFi function to send the collected data to Adafruit IO, which is a cloud service to display data in different dashboards.[15]

Previous Year's Research Summary

This project's goal is to determine the optimal seating position for short-statured drivers to reduce injury severity in a frontal collision.

In LS-DYNA, an automotive sled model was built. This model was an automobile "neutral" with a seat, seatbelt, airbag, and steering column, with no other structural safety elements included in order to focus on the interaction between dummy positioning and the aforementioned elements. This model also included the impact (40 mph), type of collision (frontal), the airbag being deployed upon impact, and an adjustable driver position of a driver wearing a seatbelt that adjusts to their position. 40 mph was chosen as it is used in standard collision tests performed by the National Institute of Highway Safety. For initial simulations, a 50th percentile FE (Finite Element) Rigid Dummy was used.

Next, real human participants were measured in order to get a pool of actual seating positions. Length measurements were taken from the line of sight (eye to windshield), nose to top of steering wheel, middle of chest to middle of steering wheel, knee to bottom of steering wheel, and knee to bottom of dashboard, as well as the pelvic angle. The human data was then used to set the positioning for the VATD in the crash by rotating or translating the limbs or entire body along the X, Y, or Z axes. The sled simulation was then run.

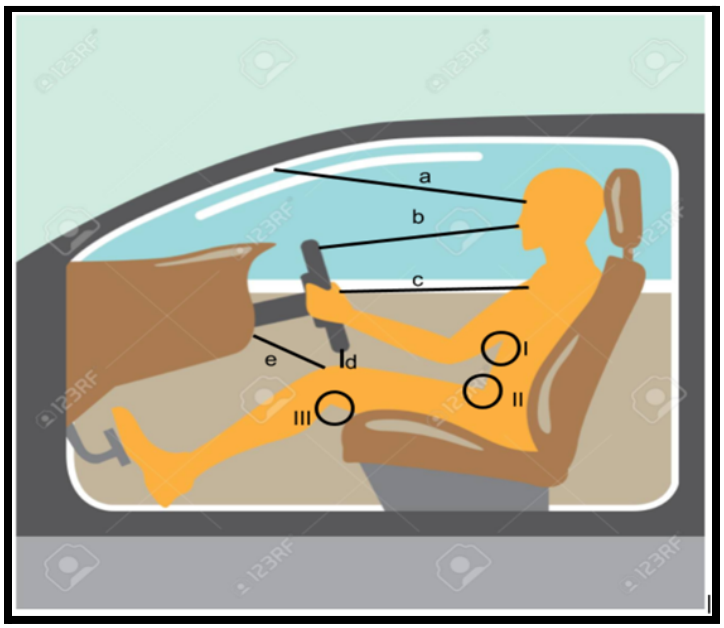
The mathematical outputs of the simulation were analyzed in post-processing in order to determine chest deflection, maximum chest acceleration, Chest Severity Index, head acceleration, and Head Injury Criteria. This data was compiled into a composite injury score, a method developed to classify total injury score by adding up points from each injury output, wherein the primary indicator is chest deflection and the least important are chest acceleration and CSI.

After completing the simulations and establishing the CIS, it was found that the shortest people (based on percentile and height) had the highest composite injury scores (Composite Injury Score Plot). Participant #15, while a seemingly higher percentile (because of age), was the overall shortest person with some of the most severe injuries, and so was selected for initial optimization. The B, C, and pelvic angle parameters were changed based on the Optimization Method Description Chart in order to find what combination minimized injury severity.

Marginal changes occurred in scenarios with bracketed changes to the position along the X axis and pelvic angle. Larger increases were not considered, as they might not be feasible due to the height of the driver (ex. reaching pedals and steering wheel, and visibility). In the final scenario with "As Seated" head and chest positions and a modified pelvic angle of 112 degrees, the Composite Injury Score decreased by 50%. This change drops the CIS from a high severity to a low severity injury.

This same process was then applied to all of the short-statured drivers. A total of 145 simulations were run, and after performing optimization simulations to the other short-statured drivers, changes to the pelvic angle, specifically a pelvic angle between 112 and 120 degrees, while keeping the head distance as close as possible to the baseline, proved to have the most

significant positive change on the injury severity scores. For the drivers whose optimized positions exhibited improved injury severity scores, the overall effect was a 39% reduction in severity.



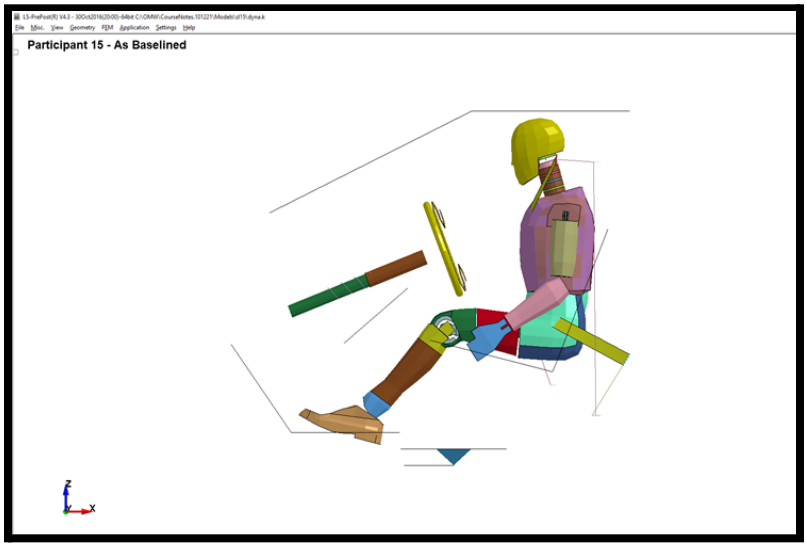
Driver Seat: Relative Position Measurements Diagram[1]

Composite Injury Score	Chest Deflection (mm)	Max Chest Acceleration (G)	Chest Severity Index	Max Head Acceleration (G)	Head Injury Criteria
1	35-45	40-65	300-650	50-65	150-300
2	46-55	66-85	651-900	66-75	301-450
3	56-65	86+	901+	76-85	451-600
4	66-75			86-95	601-699
5	76+			96+	700+

Composite Injury Score Guidelines

Demographic Data				As Baselined									As Optimized						
Participant ID	Height (cm)	Sex	Age/Height Percentile	B (mm)	C (mm)	Pelvic Angle (deg)	Chest Deflection (mm)	Max Chest Acceleration (G)	Chest Severity Index	Max Head Acceleration (G)	Head Injury Criteria	Composite Injury Score	Method	Chest Deflection (mm)	Max Chest Acceleration (G)	Chest Severity Index	Max Head Acceleration (G)	Head Injury Criteria	Composite Injury Score
1	167	M	13	445	381	109	38	133	1314	119.9	893	17	B25	37	84	692	65	666	10
4	157	F	26	280	229	108	53	52	487	66.6	561	9	B4D	45	52	451	55	499	7
6	163	F	52	440	280	120	40	44	374	61.3	581	7	F25	39	42	372	60	573	7
7	158	F	26	270	260	118	41	46	360	53.2	560	7	B4D	15	45	505	61	492	7
12	163	F	57	343	292	103	42	52	479	58.7	588	7	PAM	44	38	342	56	542	7
14	170	M	23	470	406	115	51	62	723	84.0	970	13	Sim Error						
15	150	F	24	432	286	94	83	59	555	65.8	700	14	PAM	40	42	363	59	557	7
19	158	F	28	279	248	94	42	59	547	55.9	571	7	B4D	40	44	378	59	568	7
22	155	F	17	343	279	88	49	70	838	60.0	621	11	PAM	45	43	398	63	575	7

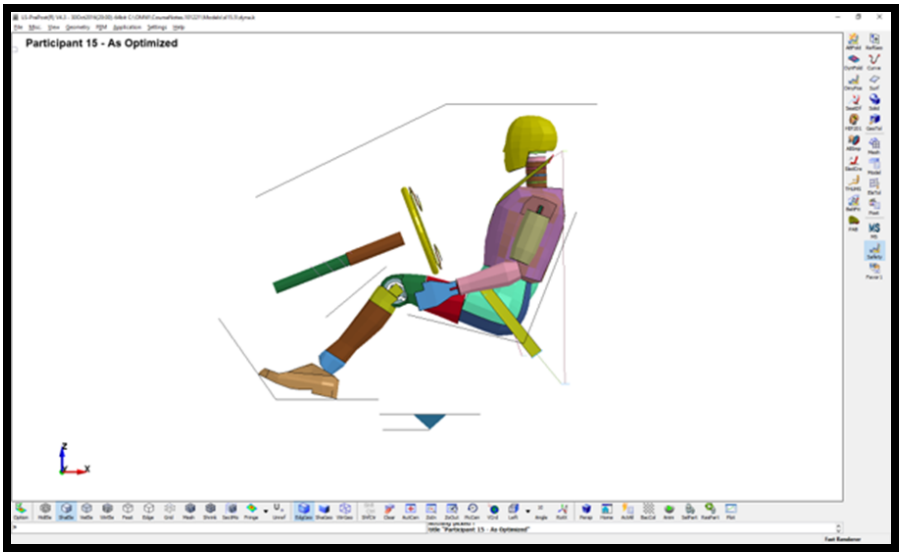
Simulation Data Results Table



Participant 15 As Baselined in LS-DYNA

Participant ID	Chest Deflection (mm)	Max Chest Acceleration (G)	Chest Severity Index	Max Head Acceleration (G)	Head Injury Criteria	Composite Injury Score
15	83	59	555	66	700	14

Participant 15 As Baselined Post-Processing Results



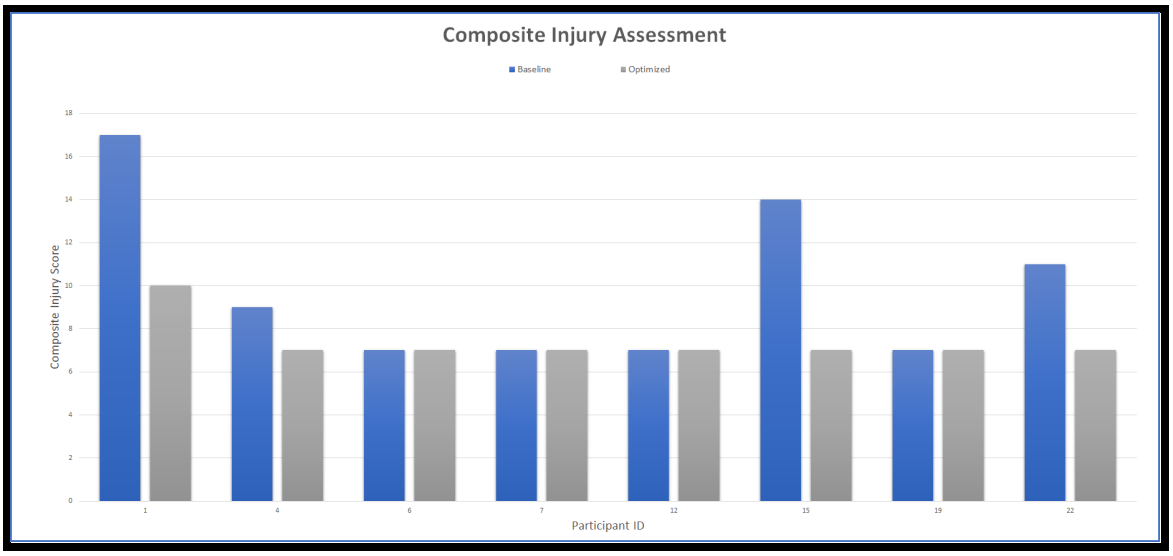
Participant 15 As Optimized in LS-DYNA

Participant ID	Chest Deflection (mm)	Max Chest Acceleration (G)	Chest Severity Index	Max Head Acceleration (G)	Head Injury Criteria	Composite Injury Score
15	40	42	363	59	557	7

Participant 15 As Optimized Post-Processing Results

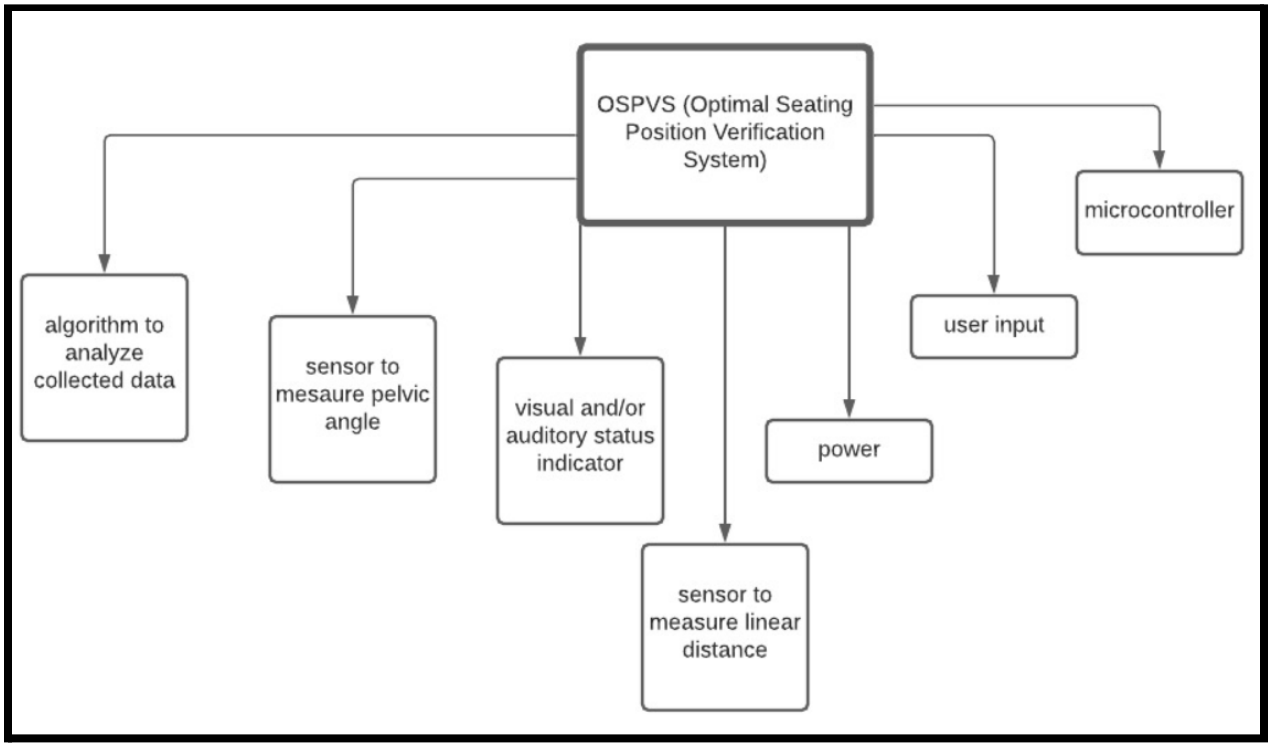
Measurement	As seated	As baselined	As optimized
B mm	432	363	432
C mm	286	286	286
Pelvic angle deg	94	94	112

Comparison of Participant 15 Measurements Table



Composite Injury Score Plot

Requirements

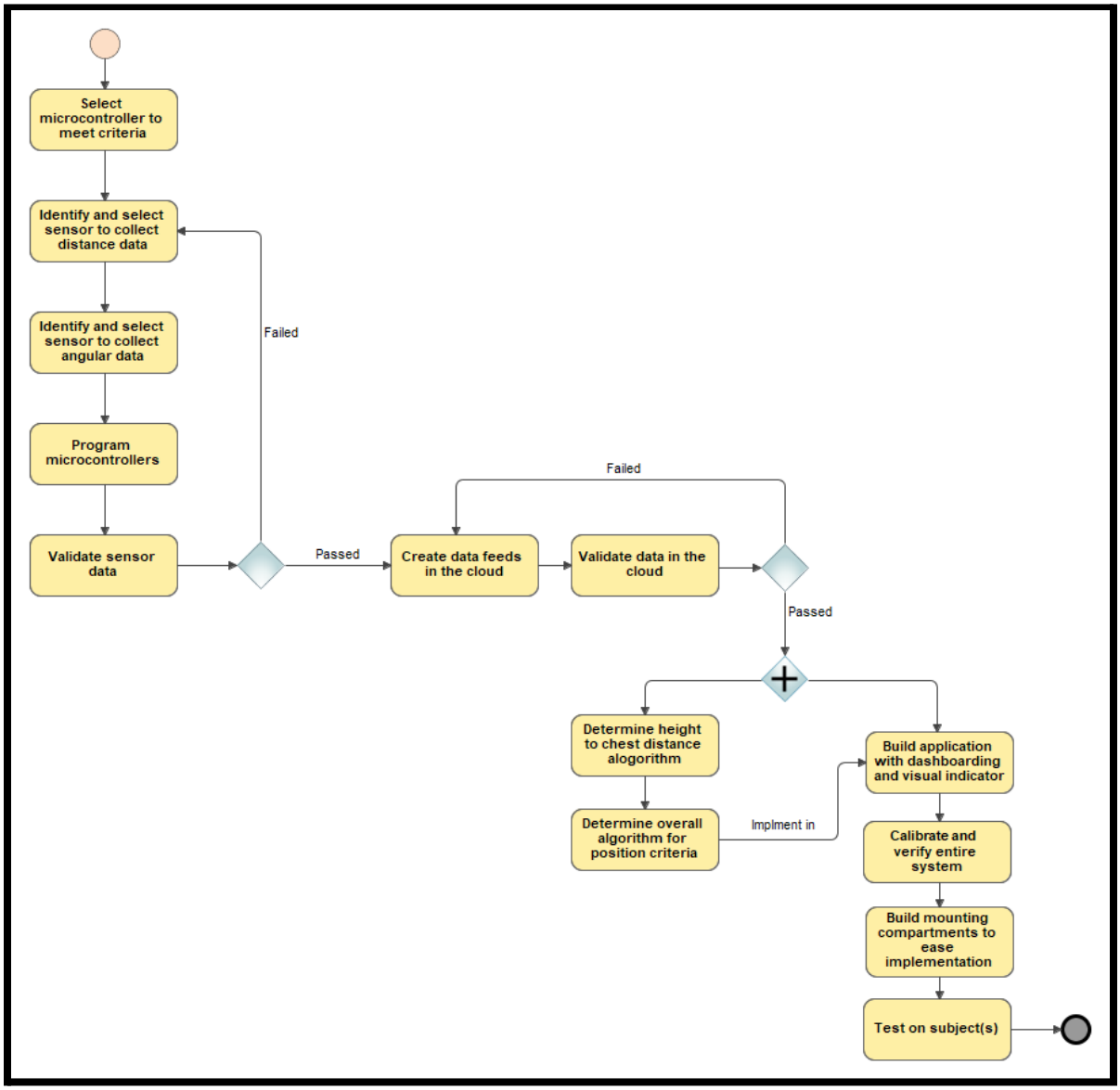


OSPVS Conceptual Block Definition Diagram generated in Lucid

Two elements requiring clarification are the sensor to measure linear distance and the algorithm. The linear distance sensor will measure chest to steering wheel distance (C) , and nose to top of steering wheel if deemed necessary and plausible. The algorithm will display accurate ranges for each measurement (chest, pelvic angle, and perhaps nose) and will display an overall

accuracy. The pelvic angle will be between 112 and 120 degrees, as determined in last year's research. The safe chest distances from last year's simulations will be plotted against height to determine a mathematical correlation between the two, and then based on the determined standard deviations, the number of standard deviations allowed as the margin of error will be determined and added to the algorithm.

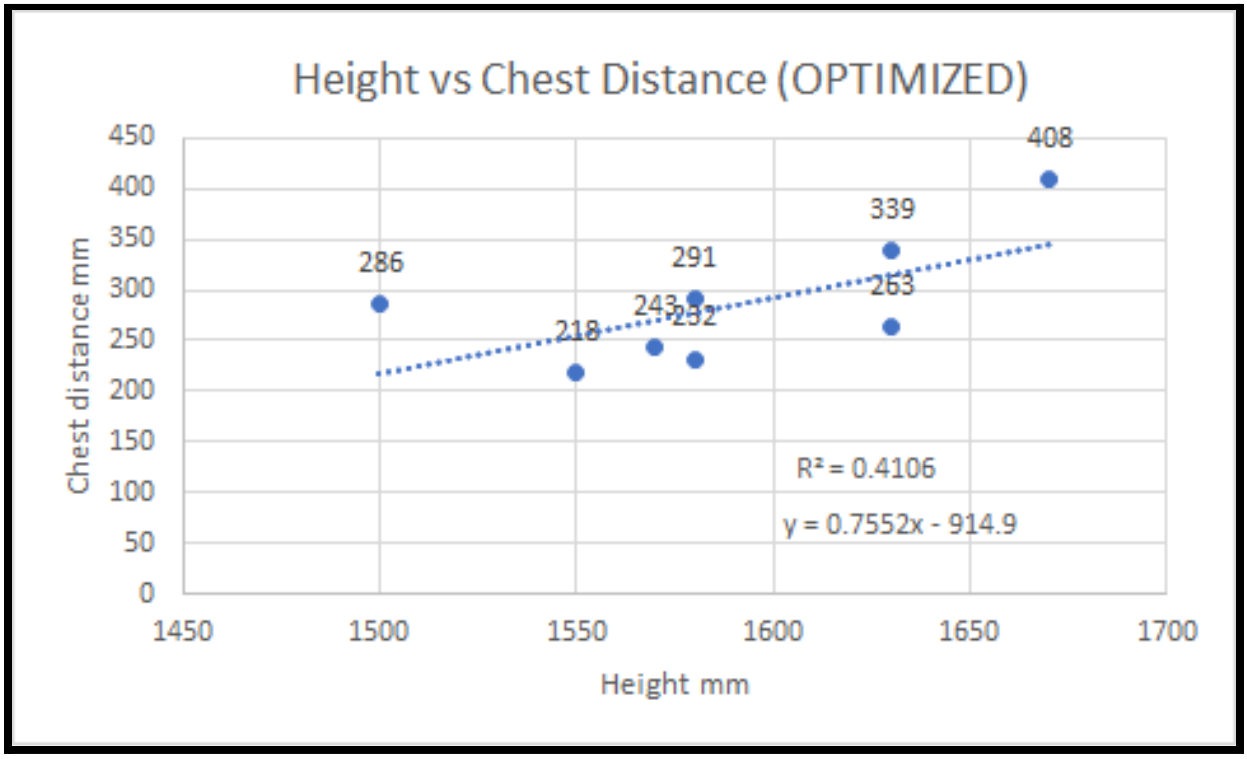
Procedure



OSPVS Procedure Flow Chart generated in Cameo Enterprise Architecture

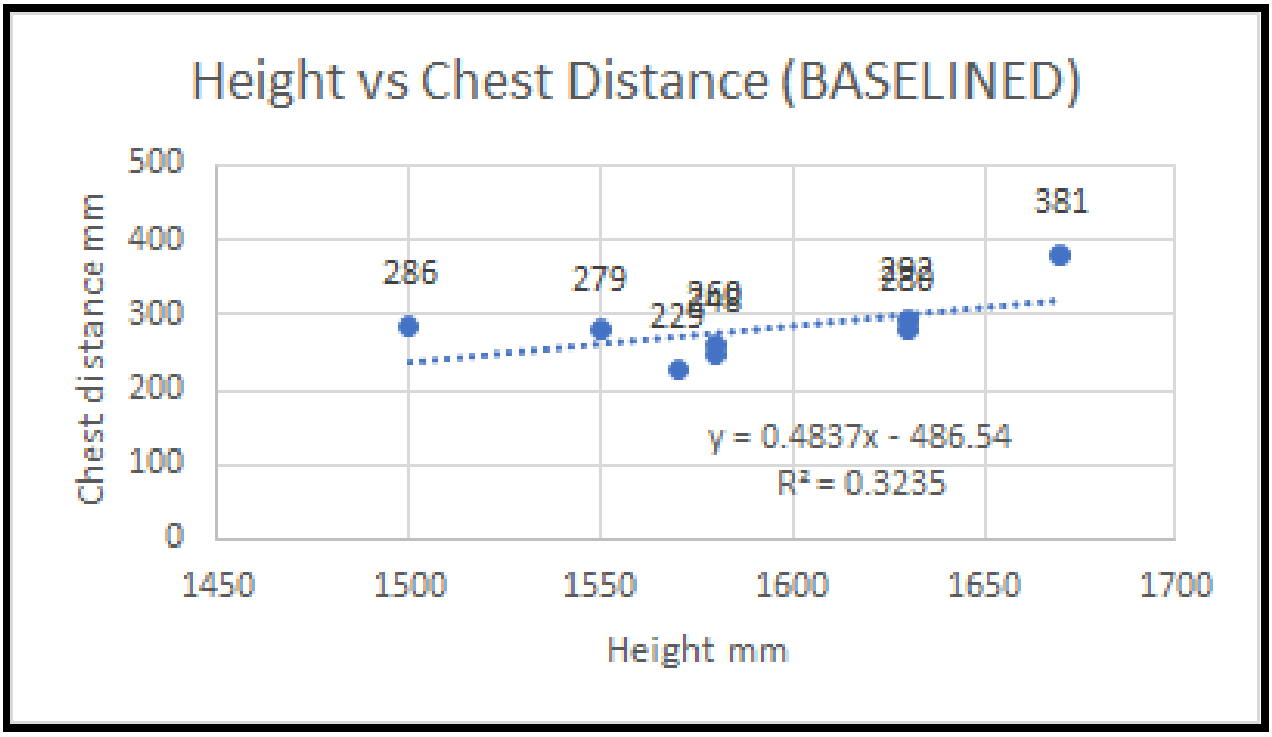
Data Results and Analysis

Height vs Chest Distance Regression



Height vs Chest Distance (Optimized) Regression generated in Excel

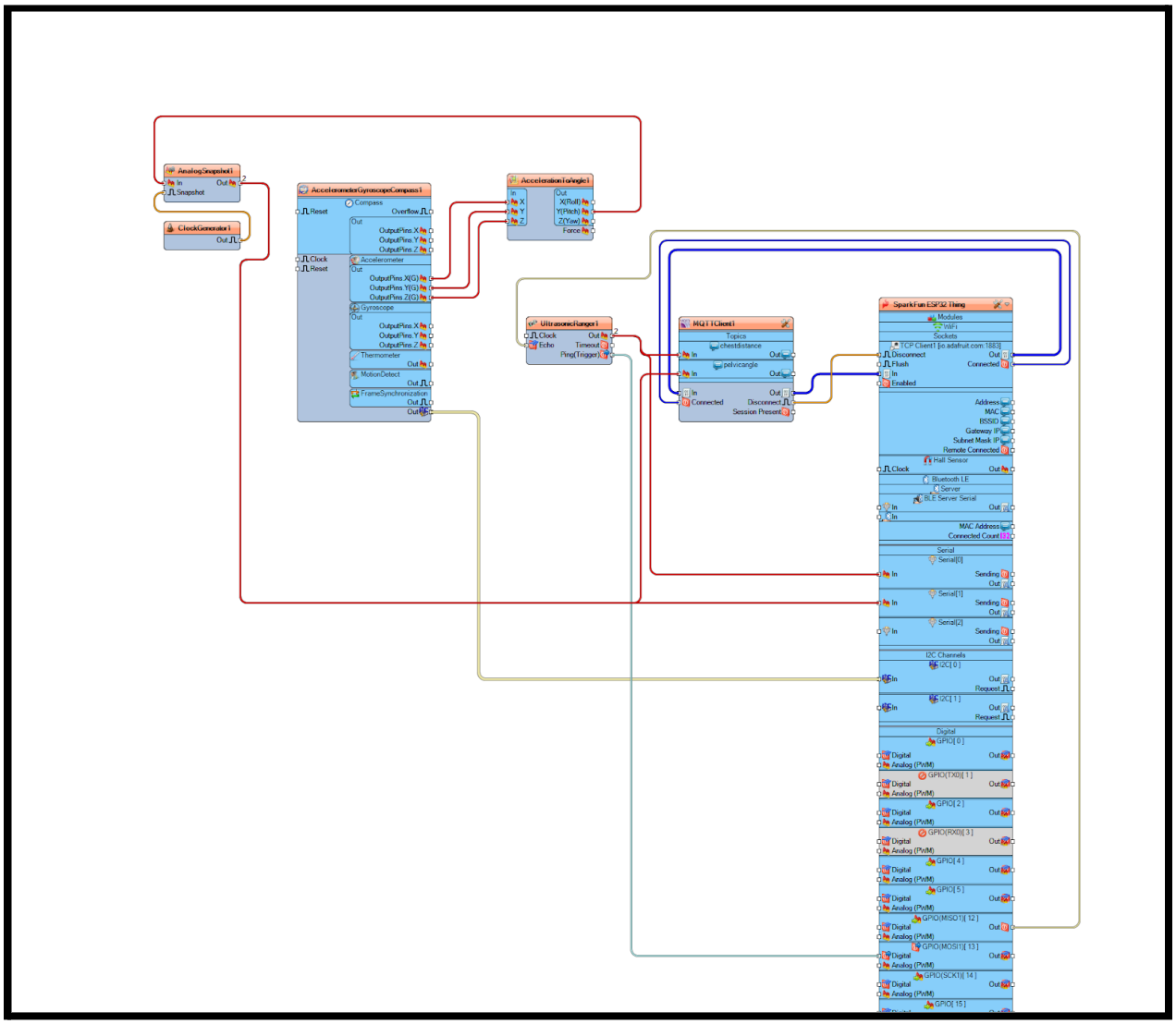
The data used in the above chart comes from the simulations run in the previous year, as seen in the Simulation Data Results Table. The chest distance was extracted from the safest positions in LS-DYNA, and plotted against height to determine an equation that could be used in the algorithm and determine its accuracy. This was done by comparing it against the same data sets of people, with the same height, but their baselined chest distance that was not optimized to safety, as seen in the Height vs Chest Distance (Baselined) chart. As the r-squared value for the optimized regression was closer to 1, the line was a better fit and shows a stronger correlation.



Height vs Chest Distance (Baselined) Regression generated in Excel

The equation from the optimized chart will be used in the algorithm. Based on the determined standard deviations, the number of standard deviations allowed as the margin of error will be determined and added to the algorithm.

Initial Iteration

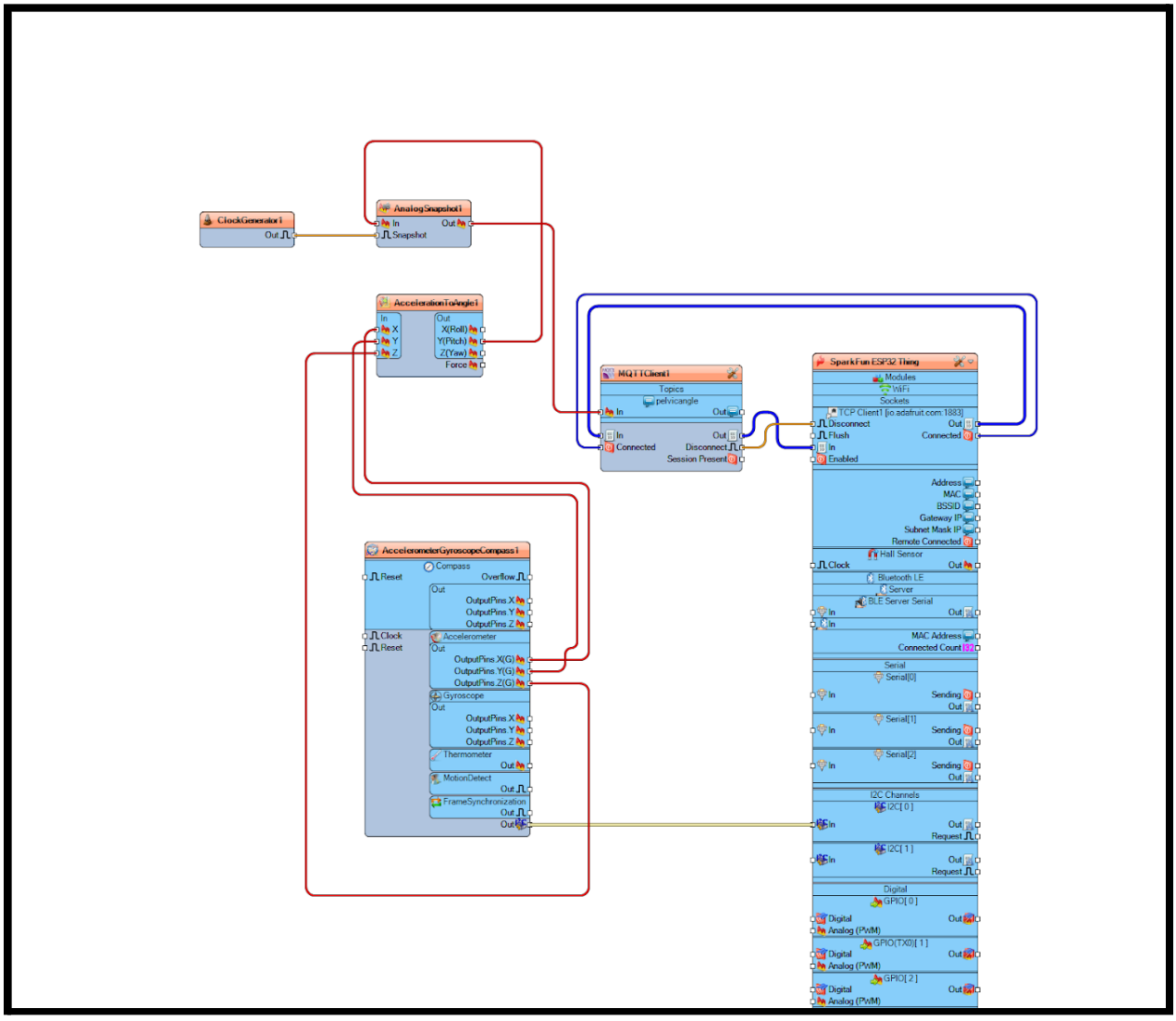


Sketch 1: Initial Visuino Sketch with One ESP32 and Two Sensors

In the initial concept for the prototype, both sensors (the MPU9250 accelerometer and the Y401 ultrasonic rangefinder) were connected to a single microcontroller, the SparkFun ESP32 Thing. The Y401 behaves just as a HC-SR04 does, but it can use either 3.3 or 5 volts of power. Sketch 1 displays the accelerometer connected to the ESP32. Acceleration data collected by the MPU9250 is then sent through the acceleration to angle function, which converts it into angular data. The analog snapshot captures data from the accelerometer and acceleration-to-angle function at a rate controlled by the clock generator. The ultrasonic range finder is also connected to the ESP32, and both the analog snapshot for angular data and the ultrasonic rangefinder send data into MQTT, which connects to two Adafruit feeds. Both were also connected to serial, so that data collection could also be viewed in Visuino for validation purposes.

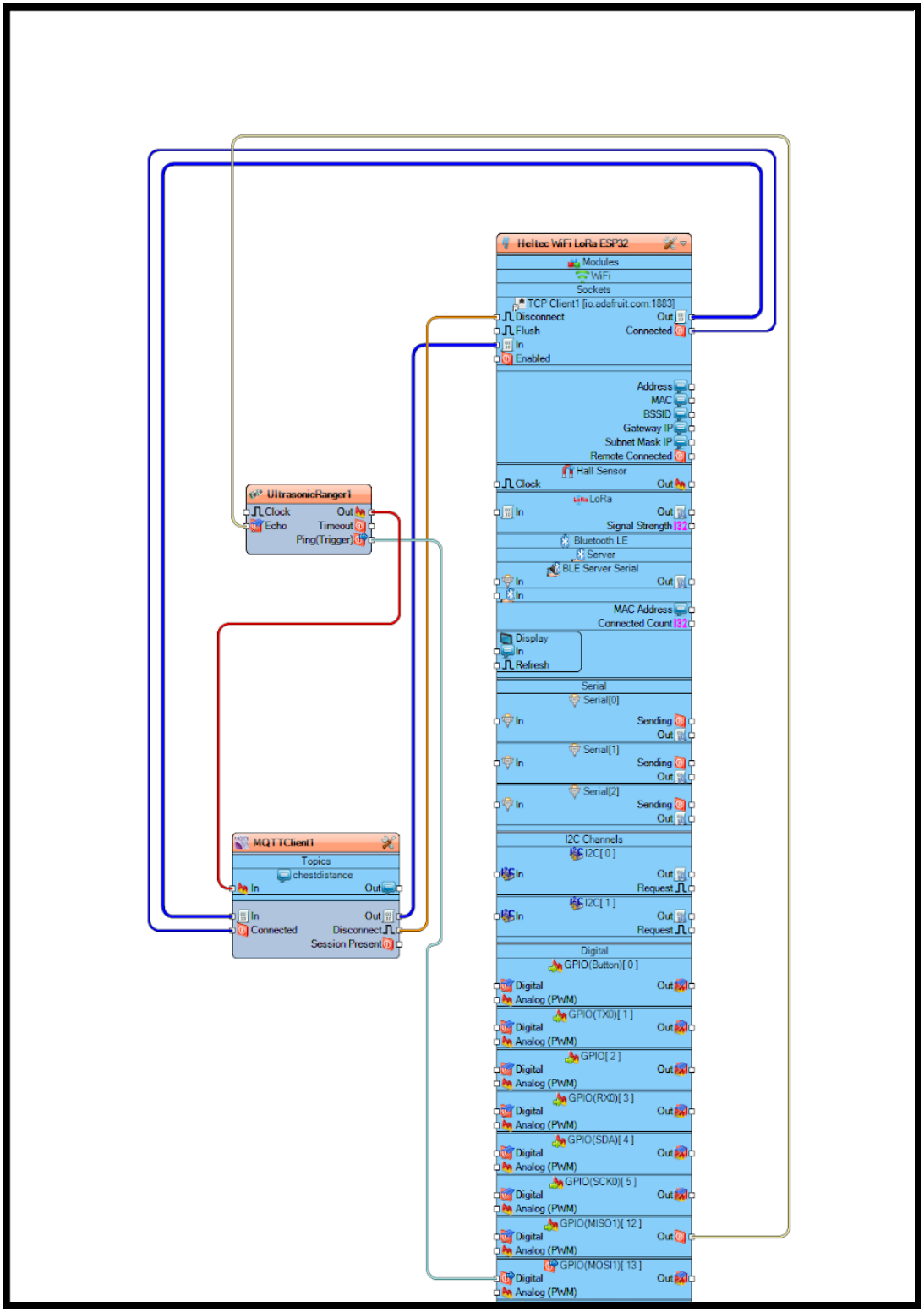
In this sketch, whichever sensor instantiated first worked to send data, and the second one wouldn't. They simply would not run together, so in the next iteration, they were decoupled and each sensor had its own microcontroller (Sketches 2 and 3).

Current Visuino Sketches



Sketch 2: Current Visuino Sketch for Pelvic Angle Collection

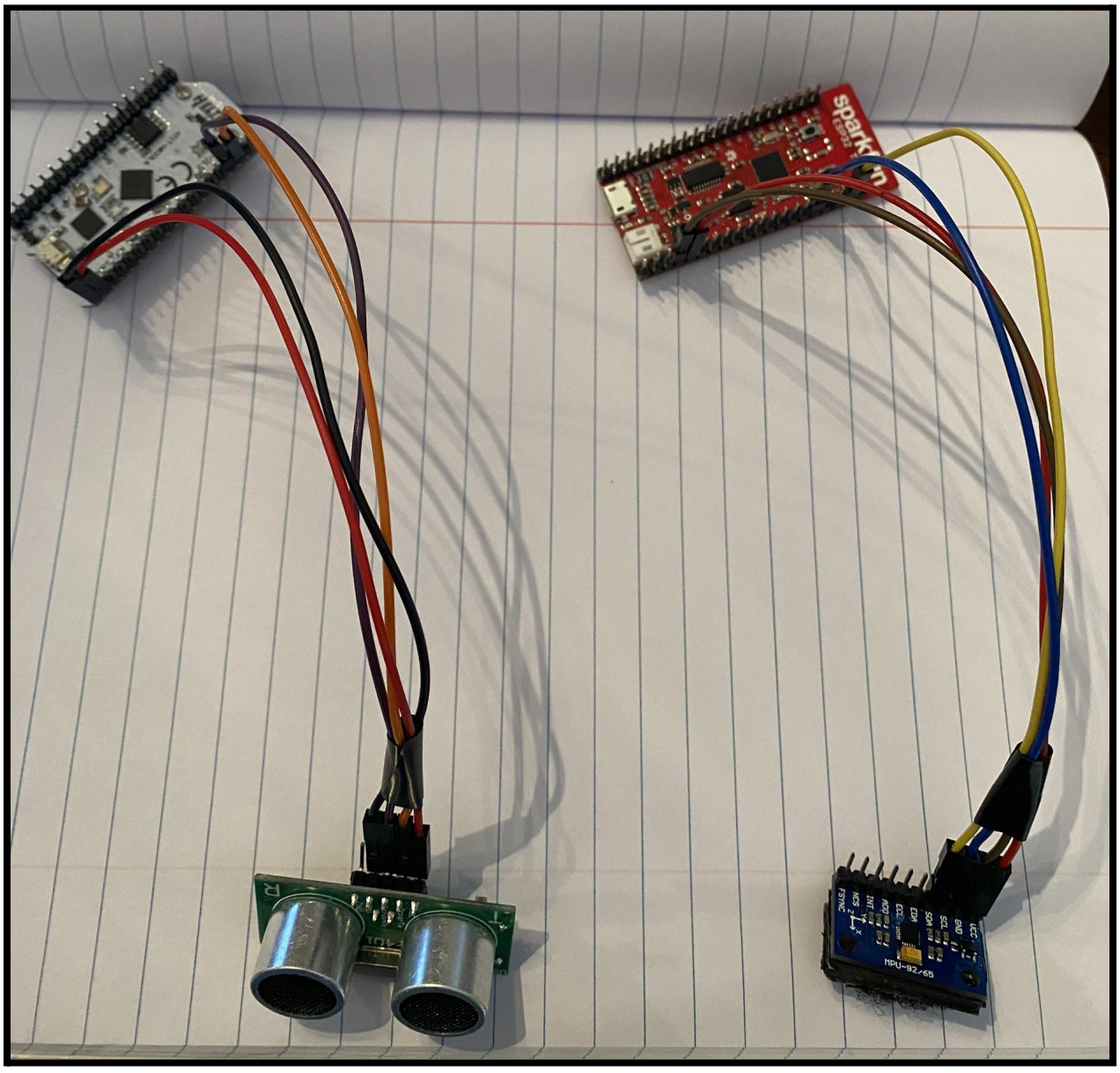
This sketch displays the accelerometer, acceleration to angle function, analog snapshot, and clock generator connected to the SparkFun ESP32 Thing and MQTT to Adafruit. The frequency of 0.2 on the clock makes it so that the analog snapshot takes measurements every 5 seconds (1/0.2).



Sketch 3: Current Visuino Sketch for Chest Distance Collection

This Visuino sketch displays the ultrasonic rangefinder connected to MQTT and the Heltec WiFi LoRa ESP32. The ultrasonic rangefinder has a Pause Time (mS) of 5000, meaning that it takes measurements every 5 seconds.

Physical Prototype



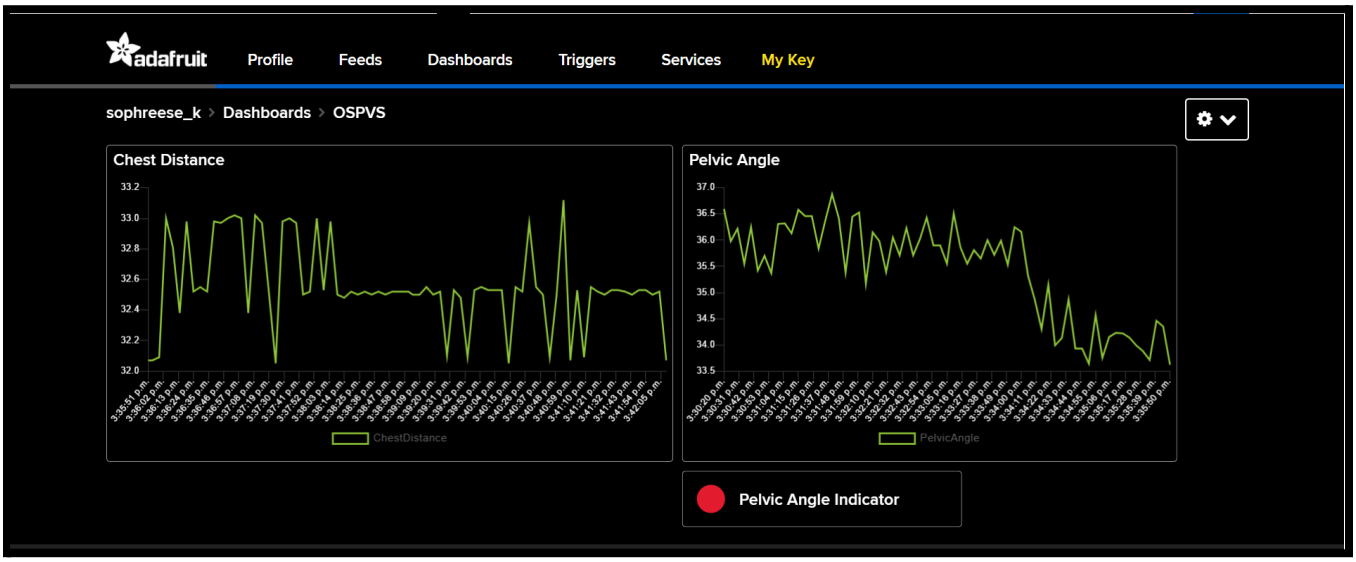
Physical Prototype: Chest Device on Left, Pelvic Angle Device on Right

The image above shows the physical implementation of the prototype. On the left is the Heltec WiFi LoRa ESP32 and the Y401 ultrasonic rangefinder, which collects chest distance. VCC is connected to 3.3V and GND is also connected to GND. The Echo pin on the sensor connects to pin #12 on the ESP32, and the Ping pin connects to pin #13. On the right is the

SparkFun ESP32 Thing and the MPU9250 accelerometer. As with the chest distance collection, it is powered by 3.3V. The SDA pin on the accelerometer connects to pin #21 on the microcontroller, and the SCL pin connects to pin #22.

Each ESP32 and sensor combination will go in a separate box. The Chest Device will have a clip to be placed in the center of the chest on the seatbelt. The Pelvic Angle Device will likely be attached to the seat itself, temporarily. The user will launch the app and be prompted to input their height and biological sex. Based on that, the algorithm will determine the optimal chest distance range. Once it has a set of parameters, the app will prompt the driver to begin adjusting their seat. As the driver adjusts their position, both indicators will show green when they are in a proper range and red if not, and when both are green, the user is in a proper position. They will then be able to store this seating position in their car (if the type of car does so) or simply leave the car seat in the position, and will be able to remove the sensors.

Adafruit Dashboard



Adafruit Dashboard Image

The image above displays the Adafruit dashboard displaying the feed for chest distance collection from the Chest Device and the pelvic angle feed from the Pelvic Angle Device. Both feeds display live data with little to no history, as this is not necessary in a real-life application of the device. The time is on the X axis and the measurement as indicated by the title is on the Y axis for both graphs. The pelvic angle indicator turns green when the pelvic angle measurement is greater than or equal to 111 degrees and less than or equal to 121 degrees. The additional degree for the margin of error is allowed due to the accuracy of the pelvic angle to 1/100th of a degree. The chest distance indicator is a work in progress.

Error Analysis

The following are possible sources of error in this project and its evaluation, as applicable to the previous year's research (simulations and optimization).

1. **Human response/personal preference:** No simulation can accurately include how a human would respond to a car crash, as likely they would know it is coming and try to take measures against that. Humans are also likely to simply have personal preference when it comes to their seating positions.
2. **Human error in measurement collection:** There is the likely possibility that error was introduced into the data through human error in data collection.
3. **Insufficient model fidelity:** Since a student edition was run, the model is limited to 10,000 finite elements, which might have limited the accuracy of the data results. However, this would only produce marginal changes and not change the statistical significance. Also, only ⅔ of the main measurements were exactly matched.

Applications and Future Plans

1. Complete the LS-DYNA simulations by getting more human data, running simulations with a 5th percentile, non-rigid dummy, and performing statistical significance tests.
2. Design the algorithm and apply it to the Chest Distance indicator in the cloud/application.
3. Plot the height vs chest distance regression again, accounting for differences in biological sex (two separate regressions).
4. Assemble the prototype, test, and validate results.
5. Create a working, if scaled down, prototype that automatically readjusts seat based on safety data.

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Acknowledgements and Statement on Outside Assistance

All graphs, charts, images, and other graphics were created by the Student Researcher unless stated otherwise. All data in the code that has been redacted is personal data such as home WiFi passwords and other keys.

What steps led you to formulate your hypothesis? My own personal experience driving as a short-statured woman gave me the idea for my research. This was then backed up by the actually completely coincidental finding of the article "The Deadly Truth About a World Built for Men," where I learned that there is data evidence backing up my own experience.

Where did you conduct the major part of your work? I conducted the majority of my research in my own home.

If you worked in an institutional setting, did you work on your project as part of a team or group? I did not work in an institutional setting, but I did receive help from Enrique Krajmalnik, who served as my mentor.

Describe what parts of the research you did on your own and what parts you received help. Enrique Krajmalnik assisted me with finding and understanding literature, learning to use LS-DYNA, working with Visuino, and running simulations. He also worked as a sounding board, helping me work through ideas and answer questions.

Is your research current or a continuation of previous research? The work presented as of today is a continuation of my previous research, as the device's development is based on the data and conclusions from the previous research.

Abstract

Short-statured drivers are at a higher risk of significant injury in car crashes, often due to adjusted positions in the seat. This imbalance highlights the need to find the optimal seating position for short-statured drivers to reduce injury severity in a frontal collision and a method for drivers to know this position. Last year, human measurements were collected and simulations were run in LS-DYNA (nonlinear finite element analysis software). After optimization, it was determined that injury severity was reduced the most when the pelvic angle was between 112 and 120 degrees, and the chest-to-middle of steering wheel and nose-to-top of steering wheel measurements were as close as possible to the baselined measurements.

This year, this data is being used to create a prototype that collects the relevant distance and angle measurements, sends them to the cloud where they are run through an algorithm to determine the optimal seating position, and displayed in a mock dashboard to inform the driver if they are in the right position or prompt readjustment. The initial prototype is working to have two ESP32 microcontrollers with one sensor (ultrasonic rangefinder/accelerometer) and is sending angle measurements to Adafruit. From last year's simulations, height has been plotted against the optimized chest measurements and produced a linear equation of chest distance=0.7552*height -914.9 with an r-squared value of 0.4106. After finding standard deviations, the number allowed as the margin of error for the chest distance measurement in the algorithm will be determined. This project will increase people's everyday safety.